

Discrete Mathematics

④

Algebraic Structures

To study a phenomenon or process of a real world, we construct a suitable mathematical model to represent it and study the properties of the model to understand the phenomenon. Often the mathematical structure of a model is presented implicitly. But in this chapter, we specify in detail some mathematical structures and develop a few basic properties of these structures, emphasizing those properties which are useful for the models under consideration. The mathematical structures are called algebras or algebraic structures. The structures mainly considered are semigroups, monoids, groups, rings and fields.

Semigroups are the simplest algebraic structures which satisfy the properties of closure and associativity. They are very

important in the theory of sequential machines, formal languages and in certain applications relating to computer arithmetic.

A monoid in addition to being a Semigroup also satisfy the identity property. Monoids are used in a number of applications but most particularly in the area of Syntactic analysis and formal languages.

Groups are monoids which also possess inverse property. The application of group theory is important in the design of fast adders and error-correcting codes.

Definition: Let S be a set and let $*$ be a binary operation on S . The operation $*$ is

(i) Commutative if $a * b = b * a$.

(ii) associative over S , if $a * (b * c) = (a * b) * c$
 $\forall a, b, c \in S$.

Ex: Let $S = \{a, b, c, d\}$ and binary operation $*$ is given by the above table.

$*$	a	b	c	d
a	a	c	d	a
b	a	b	c	d
c	a	b	a	c
d	a	b	b	b

(i) the operation is not commutative, as

$$a * b = c \text{ and } b * a = a$$

(ii) The operation is not associative as

$$a * (b * c) = a * c = d$$

$$(a * b) * c = c * c = a.$$

Binary operation

A binary operation on a non-empty set S is just a function $*$: $S \times S \rightarrow S$. So for each (a, b)

in $S \times S$, we associate an element $*$ (a, b) of S .

we shall denote $*$ (a, b) by $a * b$.

If S is a non-empty set with a binary operation $*$ then S is said to be closed under $*$.

Definition ① Let A be an algebra with a underlying set G and $*$ a binary operation on G . $(G, *)$ is called a semigroup if the following two conditions are satisfied.

1. G is closed with respect to $*$.
2. $*$ is an associative operation

Definition ②: Let $(G, *)$ be an algebraic system, where $*$ is a binary operation on G . $(G, *)$ is called a monoid if the following conditions are satisfied.

1. G is closed w.r.t $*$.
2. $*$ is an associative operation
3. There exist an identity element $e \in G$ for the operation $*$. i.e. for any $x \in G$,
$$e * x = x * e = x.$$

Definition ③: Let $(G, *)$ be an algebraic system, where $*$ is a binary operation on G . Then $(G, *)$ is called a group if the following conditions are satisfied

1. G is closed w.r.t $*$.
2. $*$ is an associative operation.
3. There exist an identity element $e \in G$ for operation $*$.
4. Each element $x \in G$ has an inverse element $x^{-1} \in G$ with respect to $*$
 i.e. $x * x^{-1} = x^{-1} * x = e$.

Definition: A group $(G, *)$ is said to be abelian if $a * b = b * a$ for all $a, b \in G$.
 If $a * b \neq b * a \forall a, b \in G$, then G is called non-abelian group.

The binary operation in a group is either denoted by $\cdot$, called multiplication or by $+$, called addition. The pair $(G, \cdot)$ then denotes a group under multiplication while $(G, +)$ denotes a group under addition.

In $(G, \cdot)$ the inverse of an element a is written as a^{-1} , while in $(G, +)$ the inverse of a is written as $-a$.

Definition: The element x of a group G is said to be idempotent if $x^2 = x$.

Theorem: The only idempotent element in a group is the identity element.

Proof: Let $x \in G$ be an idempotent element.

$$\begin{aligned} \text{Then } x^2 &= x. \\ \Rightarrow x^{-1} \cdot x^2 &= x^{-1} \cdot x \\ \Rightarrow x^{-1} \cdot x \cdot x &= e \\ \Rightarrow e \cdot x &= e \\ \Rightarrow x &= e. \end{aligned}$$

Ex (1): Consider the set $G = \{1, -1\}$ and let the binary operation defined on G be the ordinary multiplication of real numbers, then $(G, \cdot)$ is a group.

Sol.: Cayley table

$\cdot$	1	-1
1	1	-1
-1	-1	1

(i) G is closed under $\cdot$.

(ii) Identity element $= 1$.

(iii) G is associative

(iv) Inverses exist $(1)^{-1} = 1$ and $(-1)^{-1} = -1$.

$\therefore (G, \cdot)$ is a group.

Ex (2): Let $G = \{1, -1, i, -i\}$ under $\cdot$, then
Show that $(G, \cdot)$ be a group.

Sol: Cayley table since $i^2 = -1$

$\cdot$	1	-1	i	-i
1	1	-1	i	-i
-1	-1	1	-i	i
i	i	-i	-1	1
-i	-i	i	1	-1

(i) G is closed under $\cdot$.

(ii) Identity element 1.

(iii) Associative law holds, Take any three element 1, -1, i

$$a \cdot (b \cdot c) = 1 \cdot (-1 \cdot i) = 1 \cdot (-i) = -i$$

$$(a \cdot b) \cdot c = (1 \cdot -1) \cdot i = (-1) \cdot i = -i$$

(iv) Inverses exist-

$$(1)^{-1} = 1, (-1)^{-1} = -1, (i)^{-1} = -i, (-i)^{-1} = i$$

$\therefore (G, \cdot)$ is a group.

Ex (3): Let $G = \{\pm 1, \pm i, \pm j, \pm k\}$, where the symbols satisfy the relations

$$\begin{aligned} i \cdot j &= k, \quad j \cdot k = i, \quad k \cdot i = j \\ j \cdot i &= -k, \quad k \cdot j = -i, \quad i \cdot k = -j \quad \text{and} \\ i^2 &= j^2 = k^2 = -1, \quad \text{then } (G, \cdot) \text{ be a group.} \end{aligned}$$

Sol. Cayley table:

$\cdot$	1	-1	i	-i	j	-j	k	-k
1	1	-1	i	-i	j	-j	k	-k
-1	-1	1	-i	i	-j	j	-k	k
i	i	-i	-1	1	k	-k	-j	j
-i	-i	i	1	-1	-k	k	j	-j
j	j	-j	-k	k	-1	1	i	-i
-j	-j	j	k	-k	1	-1	-i	i
k	k	-k	j	-j	-i	i	-1	1
-k	-k	k	-j	j	i	-i	1	-1

(i) $(G, \cdot)$ is closed.

(ii) Identity element 1

(iii) Associative (take any 3 elements) i, j, k

$$\begin{aligned} a \cdot (b \cdot c) &= i(j \cdot k) = i(i) = i^2 = -1 \\ (a \cdot b) \cdot c &= (i \cdot j) \cdot k = k(k) = k^2 = -1 \end{aligned}$$

$\Rightarrow$ associative

(iv) Inverse:

$$\begin{aligned} (1)^{-1} &= 1, \quad (-1)^{-1} = -1, \quad (i)^{-1} = -i, \quad (-i)^{-1} = i, \quad (j)^{-1} = -j, \quad (-j)^{-1} = j \\ &\Rightarrow (G, \cdot) \text{ gp} \quad (k)^{-1} = -k, \quad (-k)^{-1} = k \quad (8) \end{aligned}$$

Ex: A

The Pairs $(\mathbb{Z}, +)$, $(\mathbb{Q}, +)$, $(\mathbb{R}, +)$, and $(\mathbb{C}, +)$, where $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$ and $\mathbb{C}$ are set of integers, rational numbers, real numbers and complex numbers respectively and $+$ denotes addition in them are all groups.

Ex (4): Let $G_5 = \{\bar{0}, \bar{1}, \bar{2}, \bar{3}, \bar{4}\}$ be the set of modulo 5. Then $(G_5, +)$ is a group.

Sol:-

(i) $(G_5, +)$ is closed.

(ii) Identity element: $\bar{0}$

(iii) Associative (take any 3)
 $\bar{2}, \bar{3}$ and $\bar{4}$

$$(a+b)+c = (\bar{2}+\bar{3})+\bar{4} = (\bar{0}+\bar{4}) = \bar{4}$$

$$a+(b+c) = \bar{2}+(\bar{3}+\bar{4}) = \bar{2}+(\bar{2}) = \bar{4}$$

$\Rightarrow$ associative

(iv) Inverses: $(\bar{0})^{-1} = \bar{0}, (\bar{1})^{-1} = \bar{4}, (\bar{2})^{-1} = \bar{3}, (\bar{4})^{-1} = \bar{1}$
 $\Rightarrow (G_5, +)$ is a group.

Cayley Table

+	$\bar{0}$	$\bar{1}$	$\bar{2}$	$\bar{3}$	$\bar{4}$
$\bar{0}$	$\bar{0}$	$\bar{1}$	$\bar{2}$	$\bar{3}$	$\bar{4}$
$\bar{1}$	$\bar{1}$	$\bar{2}$	$\bar{3}$	$\bar{4}$	$\bar{0}$
$\bar{2}$	$\bar{2}$	$\bar{3}$	$\bar{4}$	$\bar{0}$	$\bar{1}$
$\bar{3}$	$\bar{3}$	$\bar{4}$	$\bar{0}$	$\bar{1}$	$\bar{2}$
$\bar{4}$	$\bar{4}$	$\bar{0}$	$\bar{1}$	$\bar{2}$	$\bar{3}$

Ex(5): Let $\mathbb{Z}_5 = \{\bar{1}, \bar{2}, \bar{3}, \bar{4}\}$ be the set of non-zero residue classes mod 5. Then $(\mathbb{Z}_5, \cdot)$ is a group.

Sol.

(i) $(\mathbb{Z}_5, \cdot)$ is closed.

(ii) Identity element: $\bar{1}$

$\cdot$	$\bar{1}$	$\bar{2}$	$\bar{3}$	$\bar{4}$
$\bar{1}$	$\bar{1}$	$\bar{2}$	$\bar{3}$	$\bar{4}$
$\bar{2}$	$\bar{2}$	$\bar{4}$	$\bar{1}$	$\bar{3}$
$\bar{3}$	$\bar{3}$	$\bar{1}$	$\bar{4}$	$\bar{2}$
$\bar{4}$	$\bar{4}$	$\bar{3}$	$\bar{2}$	$\bar{1}$

(iii) It is associative

$$(a \cdot b) \cdot c = (\bar{2} \cdot \bar{3}) \cdot \bar{4} = \bar{1} \cdot \bar{4} = \bar{4}$$

$$a \cdot (b \cdot c) = \bar{2} \cdot (\bar{3} \cdot \bar{4}) = \bar{2} \cdot (\bar{2}) = \bar{4}$$

$\Rightarrow$ associative

(iv) Inverses: $(\bar{1})^{-1} = \bar{1}, (\bar{2})^{-1} = \bar{3}, (\bar{3})^{-1} = \bar{2}, (\bar{4})^{-1} = \bar{4}$

$\therefore (\mathbb{Z}_5, \cdot)$ is a group.

Properties of groups:

① Theorem: (Cancellation laws).

For any three elements a, b, c in a group G .

(i) $ab = ac \Rightarrow b = c$ (Left Cancellation law)

(ii) $ba = ca \Rightarrow b = c$ (Right Cancellation law)

Proof: For a, b, c in G .

$$ab = ac \Rightarrow a^{-1}(ab) = a^{-1}(ac)$$

$$\Rightarrow (a^{-1}a)b = (a^{-1}a)c \quad \text{Using associative law}$$

$$\Rightarrow eb = ec$$

$$\Rightarrow b = c.$$

$\Rightarrow$ Left Cancellation law verified.

Similarly, for the right Cancellation law.

② Theorem (Solution of linear equations)

For any two elements a, b in a group G ,
the equations $ax = b$ & $xa = b$
have unique solution $x = a^{-1}b$.

③ For any $a, b \in G$, then
 $(ab)^{-1} = b^{-1}a^{-1}$.

Proof: This follows from

$$\begin{aligned}(ab)(b^{-1}a^{-1}) &= a(b b^{-1})a^{-1} \\ &= a(e)a^{-1} \\ &= \underline{\underline{a a^{-1} = e}}\end{aligned}$$

By associativity

Definition: The number of elements in a group G is called the order of G and is denoted by $|G|$. A group G is said to be finite if G consists of only a finite number of elements. Otherwise G is said to be an infinite group.

The groups in Examples 1, 2, 3, 4, 5 are finite groups.
 The groups - Example A are infinite groups.

Definition: Let a be an element of a group G . A positive integer n is said to be the order of a if $a^n = e$,

Q ① The set $\mathbb{Z}$ of all integers under binary operation 'o' defined by $a \circ b = a - b$
 $\forall a, b \in \mathbb{Z}$.
 Is $(\mathbb{Z}, \circ)$ form a group.

Ans. It is not a group. Here $2, 3, 5 \in \mathbb{Z}$.

$$(2 \circ 3) \circ 5 = (2-3) \circ 5 = (-1) \circ 5 = -6$$

$$2 \circ (3 \circ 5) = 2 \circ (3-5) = 2 \circ (-2) = 2 - (-2) = 4$$

$\therefore (2 \circ 3) \circ 5 \neq 2 \circ (3 \circ 5)$. Not Associative.
 $\Rightarrow (\mathbb{Z}, \circ)$ is not a group.

Q.2: Show that the Set $G = \{1, 2, 4, 5, 7, 8\}$
 Under multiplication modulo 9 is a group.

Ans.

(i) $(G, \cdot)$ is closed

(ii) Identity element 1

(iii) Associative (Take $4, 5, 7$)

$$(4 \cdot 5) \cdot 7 = (2) \cdot 7 = 14 = 5$$

$$4 \cdot (5 \cdot 7) = 4 \cdot (8) = 32 = 5$$

$\Rightarrow$ Associative

Cayley Table

$\cdot$	1	2	4	5	7	8
1	1	2	4	5	7	8
2	2	4	8	1	5	7
4	4	8	7	2	1	5
5	5	1	2	7	8	4
7	7	5	1	8	4	2
8	8	7	5	4	2	1

Inverses:

$$(\bar{1})^{-1} = \bar{1}, (\bar{2})^{-1} = \bar{5}, (\bar{4})^{-1} = \bar{7} \\ (\bar{5})^{-1} = \bar{2}, (\bar{7})^{-1} = \bar{4}, (\bar{8})^{-1} = \bar{8}$$

$\Rightarrow (G, \cdot)$ is a group.

Q.3 Show that if a group G is such that $x \cdot x = e, \forall x \in G$, where e is the identity element of G , then G is abelian group.

Q.4: Prove that if every non-identity element of a group G is of order 2, then G is abelian group.

Ans: Let $a, b \in G$, then $a^2 = e, b^2 = e$ & $(ab)^2 = e$
 $\Rightarrow a = a^{-1}, b = b^{-1}$ & $ab = (ab)^{-1}$

Now $ab = (ab)^{-1} = b^{-1} a^{-1}$ by 2nd property
 $\Rightarrow ab = ba$
 $\Rightarrow$ abelian group